

EC3005 — Banking and Finance

Winter 2022 Examination · Paper 1 · Full Marking Scheme

Module	EC3005 — Banking and Finance
Internal Examiner	Dr. Fergal O'Connor
Institution	University College Cork
Session	Winter 2022, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions. Attempt 2 of 3 . Each question weighted equally (100 marks each, 100% of the paper).
Materials	Non-programmable calculator permitted

Coverage note. This scheme works through **all three** Section A questions (A1, A2, A3) in full, since you won't know in advance which two the exam will ask you to answer under exam conditions — study all three and pick your strongest pair on the day.

Own-figure rule. Where a question part carries forward from an earlier numerical or conceptual step, an answer that applies the correct method consistently from an earlier error still earns the marks available for the later steps. Markers (and you, self-marking) should apply this throughout the numerical parts of A3.

Consistency with other sittings. Three question parts on this paper — A1(a) risk vs. uncertainty, A2(a) primary vs. secondary markets, and A2(c) intermediary functions — cover the same theory as questions already built for the Winter 2025 sitting. The underlying explanations below are word-for-word identical to that scheme (drawn from a shared theory library), so cross-checking the two never turns up a contradiction — only the surrounding question wording and mark weightings differ where the papers themselves differ.

Source basis. Content and worked methods follow Dr. O'Connor's own EC3005 course pack and lecture notes, cross-checked against the module's core texts — Mishkin, *The Economics of Money, Banking and Financial Markets*, and Bodie, Kane & Marcus, *Investments* — for the standard treatment of any concept not spelled out in the lecture notes.

Section A · Question 1

Risk vs. uncertainty, moral hazard in debt contracts, Basel III capital buffers — 100 marks

A1(a)

[30 marks]

Discuss the difference between risk and uncertainty. Use an example in your answer.

A1(b)

[40 marks]

Discuss **three** tools that help solve the moral hazard problem in debt contracts.

A1(c)

[30 marks]

Discuss **one** of the following two capital buffers which are set out in the Basel III international regulatory framework for banks: (i) Capital conservation buffer (ii) Countercyclical capital buffer.

FULL MODEL ANSWER

The distinction (Knight, 1921): **risk** describes a situation where the possible outcomes are known and their **probabilities can be estimated** — from historical frequency, from theory, or from a well-understood random process. **Uncertainty** describes a situation where either the possible outcomes themselves, or the probabilities attached to them, **cannot be meaningfully estimated** at all.

- **Risk example:** the probability a diversified portfolio of investment-grade corporate bonds defaults in a given year can be estimated reasonably well from decades of historical default-rate data — an insurer or bank can price this and hold capital against it.
- **Uncertainty example:** the economic impact of a genuinely novel event — a first-of-its-kind pandemic, or an unprecedented geopolitical shock — has no comparable historical base rate to draw a probability distribution from; markets and models can only guess, not calculate.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Risk and uncertainty correctly distinguished on the basis of whether probabilities can be estimated AND a clear, well-chosen example given.
15-22	15-22	Correct distinction given, but the example is generic or only weakly illustrates the distinction (e.g. "the stock market is risky" without saying why that's risk rather than uncertainty).
8-14	8-14	Distinction attempted but imprecise (e.g. "uncertainty is just bigger risk") with no usable example.
0-7	0-7	Risk and uncertainty used interchangeably with no meaningful distinction drawn.

The problem: moral hazard in a debt contract arises once the loan is disbursed — the borrower has an incentive to take on riskier projects than the lender would approve of, since the borrower keeps the upside if the risky project succeeds but the lender bears most of the downside if it fails (the borrower's liability is limited to losing their own equity stake).

- **Net worth and collateral requirements:** requiring the borrower to pledge collateral, or to have meaningful equity/net worth at stake, means the borrower has more to lose personally if a risky project fails — this brings the borrower's incentives closer to the lender's own, since the borrower now bears more of the downside too.
- **Restrictive covenants:** the loan contract writes in explicit restrictions on what the borrower can do with the funds — e.g. covenants restricting the borrower from taking on additional risky ventures, requiring the collateral to be maintained, or requiring minimum levels of activity that discourage risk-shifting — directly limiting the moral hazard behaviour rather than just changing incentives.
- **Monitoring and enforcement:** the lender (or a delegated monitor acting on the lender's behalf, such as a bank) incurs the cost of periodically verifying that the borrower is actually complying with the covenants and using the funds as agreed — covenants only reduce moral hazard if breaches are actually detected and enforced, so monitoring is what makes the other two tools effective in practice.

EXAM TECHNIQUE

These three work together rather than in isolation — collateral changes the borrower's *incentives*, covenants set explicit *limits* on behaviour, and monitoring is what makes those limits *enforceable*. A strong answer notes this rather than treating the three as an unconnected list.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Three distinct tools given AND each explained with the correct mechanism by which it reduces moral hazard in a debt (not equity) contract specifically AND the borrower/lender incentive misalignment is stated at the outset.
21-30	21-30	Three tools named with broadly correct mechanisms, but one explanation is thin or generic, or the debt-specific framing (limited liability, risk-shifting) is missing.
11-20	11-20	Only two tools given, or three tools listed with little explanation of mechanism.
0-10	0-10	Fewer than two credible tools given, or the answer confuses moral hazard with adverse selection throughout.

EXAM TECHNIQUE

The question asks you to discuss **one** of the two buffers — both full answers are given below so you can prepare either, but only one is required (and marked) on the day.

Option (i) — Capital conservation buffer: an additional layer of **Common Equity Tier 1 (CET1) capital**, set at 2.5% of risk-weighted assets on top of the minimum capital requirement, that banks must hold in normal times.

Mechanism: it is not a hard minimum in the sense of triggering immediate resolution if breached — instead, if a bank's capital ratio falls into the buffer range, automatic restrictions kick in on discretionary distributions (dividends, share buybacks, discretionary staff bonuses), forcing the bank to conserve capital and rebuild the buffer through retained earnings rather than continuing to pay it out. **Purpose:** this converts a fixed minimum (which a bank has every incentive to run right up against) into a graduated buffer the bank has strong incentives to keep intact, so capital is actually available to absorb losses when a downturn arrives rather than already being paid out.

Option (ii) — Countercyclical capital buffer: an additional CET1 capital requirement, set by national macro-prudential authorities in a range of **0% to 2.5%** of risk-weighted assets, that is adjusted over the credit cycle rather than fixed. **Mechanism:** the buffer is raised (typically toward the top of the range) during periods of excessive credit growth, forcing banks to hold more capital precisely when systemic risk is building up, and released during a downturn — freeing up that capital so banks can keep lending rather than deleveraging into the crisis and worsening it. **Purpose:** unlike the (roughly) fixed capital conservation buffer, this buffer is explicitly a macro-prudential tool (see Q. A2(b) of the Winter 2025 sitting scheme for the macro/micro-prudential distinction) — it targets the build-up of system-wide risk over time, not just an individual bank's static soundness.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	The chosen buffer's size/basis correctly stated (% of risk-weighted assets) AND the trigger mechanism correctly explained (restrictions on distributions for the conservation buffer; cyclical adjustment for the countercyclical buffer) AND the regulatory purpose clearly stated in its own right, not just restated from the definition.
15-22	15-22	Buffer correctly identified and roughly sized, with the mechanism described but not fully precise (e.g. "the bank has to hold more capital" without explaining what actually happens if the buffer is breached or how it is set).
8-14	8-14	Buffer named with only a vague description of what it does, no reference to the trigger mechanism or the size.
0-7	0-7	Neither buffer meaningfully described, or the two buffers are conflated with each other or with the minimum capital requirement itself.

Question A1 Total: 100 Marks

Section A · Question 2

Primary vs. secondary markets, maturity transformation, financial intermediation and growth — 100 marks

A2(a)

[30 marks]

Financial markets can be classified as primary and secondary markets. Outline the difference between primary and secondary markets.

A2(b)

[30 marks]

Explain how banks carry out Maturity Transformation. Why are they able to do it?

A2(c)

[40 marks]

Financial intermediaries and markets can alleviate market frictions and contribute to economic growth by performing the following four functions: (i) Producing information and allocating capital; (ii) Monitoring firms and exerting corporate governance; (iii) Mobilising/pooling savings; (iv) Easing exchange of goods and services. Discuss **any two** of the above functions and comment on how each function may contribute to economic growth.

FULL MODEL ANSWER

Primary market: the market in which a security is sold **for the first time** — the issuer (a company issuing new shares, or a government issuing new bonds) receives the proceeds directly. An IPO and a government bond auction are both primary-market transactions.

Secondary market: the market in which securities that have **already been issued** are bought and sold between investors. The issuer is not a party to the trade and receives no funds from it — the New York Stock Exchange and the Irish Stock Exchange (Euronext Dublin) are secondary markets.

Why this matters: the two are functionally linked, not just definitionally different. A deep, liquid secondary market is what makes investors willing to buy in the primary market in the first place — an investor who could not resell a bond or share easily would demand a much higher return to compensate for that illiquidity, raising the cost of raising finance for issuers. Secondary-market liquidity is therefore a precondition for a well-functioning primary market, not a separate phenomenon.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correct distinction stated explicitly in terms of who receives the funds (issuer vs. no one — trade is between investors) AND at least one concrete example of each market AND the liquidity link between the two explained.
15-22	15-22	Correct distinction with at least one example, but the liquidity/pricing link

between primary and secondary markets is not explained, or is asserted without reasoning.

8-14	8-14	Only one of the two markets is correctly defined, or the distinction is stated vaguely (e.g. "new vs old shares") without reference to who receives the funds.
0-7	0-7	No meaningful distinction made, or primary/secondary is confused with an unrelated classification (e.g. money market vs. capital market).

Mechanism: banks fund themselves largely with **short-maturity liabilities** — demand deposits that can be withdrawn at any time — while holding **long-maturity assets** — mortgages and term loans that repay only over years or decades. This mismatch between the maturity of what a bank owes and the maturity of what it owns is **maturity transformation**: the bank is, in effect, converting short-term, liquid funding into long-term, illiquid lending.

Maturity Transformation: Liability Maturity vs. Asset Maturity

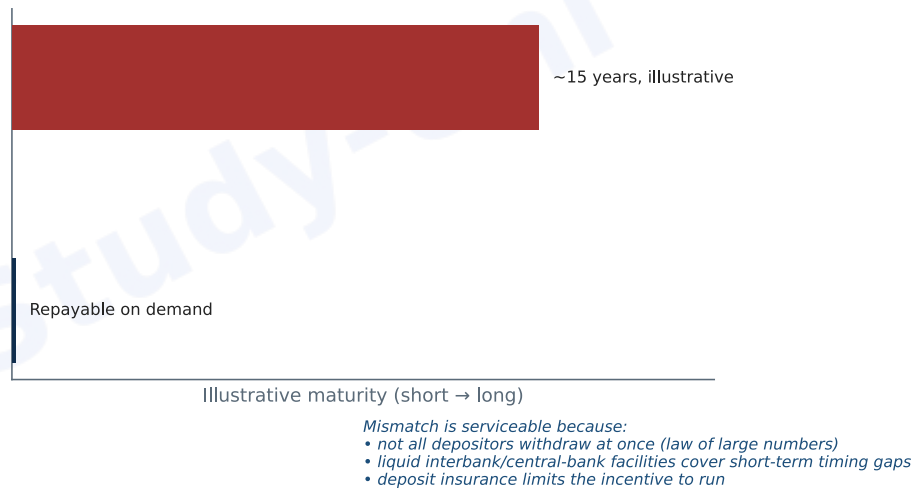


Fig. 1 — Banks fund long-maturity assets (loans, mortgages) with short-maturity liabilities (demand deposits), a mismatch that is serviceable only because not all depositors withdraw simultaneously.

Why banks are able to do it: three things make the mismatch serviceable rather than a permanent liquidity crisis: (1) the **law of large numbers** — with a large number of depositors whose individual withdrawal timing is largely independent, only a predictable fraction of total deposits is withdrawn on any given day, so the bank only needs to keep a fraction of deposits liquid, not the whole amount; (2) access to **liquid interbank and central-bank facilities** to cover short-term funding gaps that do arise; and (3) **deposit insurance**, which removes most depositors' incentive to withdraw purely out of fear that other depositors might withdraw first — without it, the mismatch would be far more fragile to a self-fulfilling bank run.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies the short-liability/long-asset mismatch AND explains at least two of the three reasons banks can sustain it (law of large numbers, liquidity facilities, deposit insurance) AND uses the term "maturity transformation" or an equivalent precise description.
15-22	15-22	Correctly identifies the mismatch, but only one reason it is sustainable is given, or the reasons given are vague ("banks manage it carefully").
8-14	8-14	Vague reference to banks "borrowing short and lending long" without explanation of why this is possible without constant liquidity crises.
0-7	0-7	No credible description of maturity transformation, or it is confused with

EXAM TECHNIQUE

The question asks for **any two** of the four functions listed — answering all four does not earn extra marks and risks running out of time relative to the depth expected per function. This answer covers (i) producing information and allocating capital and (ii) monitoring firms and exerting corporate governance; either of the other two pairings is equally valid if argued with the same depth.

Producing information and allocating capital: individual savers face high costs in researching which borrowers are creditworthy — this is exactly the kind of task with large fixed costs and economies of scale, which is why it is delegated to specialist intermediaries rather than done by each saver separately. Banks and other intermediaries develop the expertise to screen borrowers before lending (**reducing adverse selection**) and monitor them afterwards. **Growth link:** by channelling savings toward the projects with the highest genuine returns rather than the projects that merely look best on paper, intermediaries raise the average productivity of the economy's capital stock — capital is allocated to its most productive use rather than spread inefficiently across borrowers who cannot be properly screened by individual savers.

Monitoring firms and exerting corporate governance: once a loan or investment is made, the intermediary has an ongoing incentive to monitor the firm's use of funds (**reducing moral hazard**) — a bank can attach loan covenants, and equity intermediaries such as venture capital and private equity funds routinely take board seats. **Growth link:** this disciplines management toward value-maximising decisions and away from empire-building or shirking, which — aggregated across the whole economy — raises the return firms generate on invested capital and therefore the rate at which the capital stock itself grows. (See also Q. A1(b) above, where restrictive covenants and monitoring are examined as tools against moral hazard specifically in debt contracts.)

Mark Allocation — Part (c) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Both chosen functions correctly defined AND a specific, non-generic growth mechanism given for each (not just "this helps the economy") AND the underlying market friction (adverse selection/moral hazard/coordination cost) each function addresses is named.
21-30	21-30	Both functions correctly defined with a growth link stated, but the growth link is generic/asserted rather than mechanistically explained, or the underlying friction is not identified.
11-20	11-20	Only one function is developed to the standard above; the second is defined but not linked to growth, or is materially incomplete.
0-10	0-10	Neither function is developed beyond a one-line definition, or fewer than two functions are meaningfully addressed.

Question A2 Total: 100 Marks

Section A · Question 3

Bond pricing at par, the risk-free asset, finite-horizon dividend valuation — 100 marks

A3(a)

[30 marks]

Laoi Ltd. has issued a bond with the following characteristics: par value: €1,000. Time to maturity: 2 years. Coupon rate: 9 percent. The coupon payments are semi-annual. Calculate the price of this bond if the Yield to Maturity is 9 percent.

A3(b)

[40 marks]

Explain what is meant by the term "Risk-Free Asset"? Use examples in your answer.

A3(c)

[30 marks]

Miller Corporation has an odd dividend policy. The company has just paid a dividend of €15 per share and has announced that it will increase the dividend by €2 per share for each of the next 5 years, and then never pay another dividend. If you require a 11 percent return on the company's stock, how much will you pay for a share today?

FULL MODEL ANSWER

Setting up the problem: with semi-annual coupons, a 2-year bond again has **$n = 4$ periods**. Here the coupon rate and the YTM are both 9%, so the **semi-annual coupon** is $9\% \times €1,000 \div 2 = \text{€45 per period}$ and the **semi-annual discount rate** is $9\% \div 2 = \text{4.5% per period}$ — coupon rate and discount rate are identical here, which is the key feature of this question.

Period (t)	Cash flow (€)	Discount factor (1.045) ^{-t}	PV of cash flow (€)
1	45.00	0.9569	43.06
2	45.00	0.9157	41.21
3	45.00	0.8763	39.43
4	1,045.00 (coupon + par)	0.8386	876.30
Bond price			€1,000.00

Result: the bond prices **exactly at par** (€1,000.00). This is not a coincidence: whenever the **coupon rate equals the YTM**, a bond always prices at par, regardless of maturity or compounding frequency — the coupon is exactly compensating the investor at the rate the market currently demands, so there

is no premium or discount to apply. (Compare Q. A3(a) of the Winter 2025 sitting scheme, where a 6% coupon against a 5% YTM produces a premium — the same formula, the opposite relationship between coupon and YTM.)

EXAM TECHNIQUE

Recognising that coupon rate = YTM $\Rightarrow$ price = par is a useful sanity check on any bond-pricing question — if your final worked answer for a bond priced at coupon = YTM comes out to anything other than par value, you have made an arithmetic error somewhere in the discounting and should re-check each period's cash flow and discount factor.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly converts to 4 semi-annual periods, €45 coupon, and 4.5% discount rate AND correctly discounts all four cash flows AND arrives at €1,000.00 (or recognises directly that coupon = YTM $\Rightarrow$ price = par, with the calculation shown to confirm it).
15-22	15-22	Correct semi-annual setup (periods, coupon, rate) but an arithmetic error in the discounting that produces a result other than par, without the error being carried through consistently.
8-14	8-14	Bond-pricing formula applied on an annual basis (2 periods, €90 coupon, 9% rate) — wrong method for a semi-annual bond, but formula mechanically applied correctly and still correctly arrives at par given the same reasoning.
0-7	0-7	No coherent application of the bond-pricing formula, or par value omitted from the final period entirely.

Definition: a **risk-free asset** is an asset whose future return is known with certainty — there is no risk that the promised payment will not be made in full and on time. Formally, its return has **zero variance** and, critically, **zero covariance** with the returns of every other asset in the market — it does not just have low risk in isolation, it contributes no risk when added to a portfolio.

Why true risk-free assets are rare in practice: no privately-issued asset is truly risk-free, since any private issuer can in principle default. In practice, **short-term government debt of a fiscally strong, currency-issuing sovereign** — for example, short-dated U.S. Treasury bills, or short-dated German Bunds within the eurozone — is treated as the closest real-world approximation, since the probability of default is judged negligible over a short horizon and the issuer can, in the case of a currency-issuing government, create the currency needed to service the debt.

Why it matters: the risk-free rate is the essential building block of asset pricing — it is the baseline return investors could earn while bearing no risk at all, so it is the reference point every other required return (in the CAPM, in discounting risky cash flows, and in the YTM comparisons used elsewhere on this paper) is built up from by adding a risk premium.

EXAM TECHNIQUE

A common error is naming **any** government bond as "the" risk-free asset without qualification — government debt of a country facing genuine default risk (a lower-rated sovereign, or a country without control of its own currency and under fiscal stress) is not risk-free, and long-dated bonds of even a strong sovereign still carry interest-rate risk if sold before maturity. Precision about *which* instrument and *which* risk (default vs. interest-rate risk) is part of what separates a strong answer from a vague one.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correct definition in terms of certainty of payment AND zero variance/covariance with the market AND at least one well-chosen, correctly-qualified example (e.g. short-dated T-bills) AND explains why the risk-free rate matters as a pricing benchmark.
21-30	21-30	Correct definition with a valid example, but the zero-covariance property or the pricing-benchmark role is missing.
11-20	11-20	Vague definition ("an asset with no risk") with no discussion of variance/covariance, though a plausible example is given.
0-10	0-10	No credible definition given, or an asset with material default or interest-rate risk is presented as risk-free without qualification.

Recognising the structure: this is **not** a constant-growth or differential-growth perpetuity — the dividend grows by a **fixed euro amount** (€2), not a fixed percentage, and the company **stops paying dividends entirely** after year 5. With no dividend after year 5 and no terminal sale value given, the share is worth precisely the present value of the five dividends the company has committed to — nothing more.

Year (t)	Dividend (€)	Discount factor $(1.11)^{-t}$	PV of dividend (€)
1	17.00	0.9009	15.32
2	19.00	0.8116	15.42
3	21.00	0.7312	15.36
4	23.00	0.6587	15.15
5	25.00	0.5935	14.84
Price today			€76.08

Result: the share is worth **€76.08** today — the sum of five discounted dividend payments and nothing else, since the €15 dividend just paid (D_0) belongs to the previous owner and is not part of today's price, and there is no cash flow of any kind after year 5 to discount.

EXAM TECHNIQUE

The single most common error on this question is reaching for the constant-growth or differential-growth formulas on the exam's own formula sheet ($P_0 = \text{Div}_1 / (R - g)$, or the two-stage version) — both assume dividends grow at a **percentage** rate and continue in perpetuity (directly or via a terminal value). Neither condition holds here: growth is a flat €2 per year, and dividends stop dead after year 5. Applying either formula anyway is a conceptual error, not an arithmetic one, and the own-figure rule does not rescue an answer built on the wrong model.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies this as a finite 5-year cash-flow stream (not a growth-perpetuity) AND correctly computes all five dividends AND correctly discounts and sums them to €76.08.
15-22	15-22	Correctly identifies the finite-horizon structure and computes the five dividends correctly, but makes a discounting arithmetic error in one or two periods.
8-14	8-14	Dividends computed correctly but the constant-growth or differential-

growth perpetuity formula is wrongly applied instead of discounting the five cash flows directly.

0-7

0-7

Dividend sequence itself computed incorrectly (e.g. treating the €2 as a percentage growth rate), compounding the conceptual error.

Question A3 Total: 100 Marks